

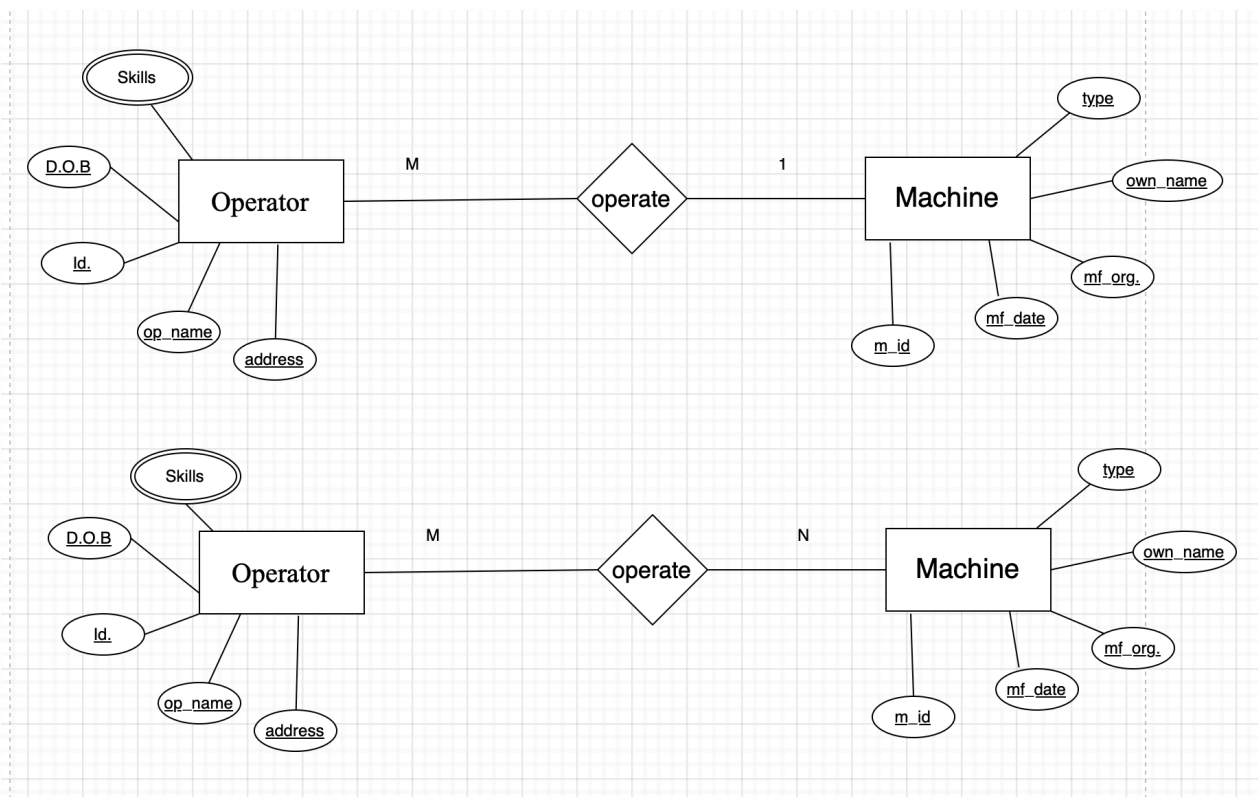
Lab 1(BCS551) – E R Diagram

Objective: To understand a given case study, notations used for ER Diagram and design the ER Diagram using a tool of following scenarios.

Exercise-1 (Basic Level)

Case Study 1:

Draw E-R diagrams to indicate the following relationships between entity set Operator and entity set Machine: - Each Machine can be operated by many Operators but each Operator can operate only one machine. An operator can operate many machine and each machine can be operated by many Operators.



Exercise-2 (Moderate Level)

Entity Relationship Model

Case Study 2:

Draw the ER diagram of a College Management System with the following details:-

- A college contains many departments
- Each department can offer any number of courses
- Many instructors can work in a department
- An instructor can work only in one department

- For each department there is a Head
- An instructor can be head of only one department
- Each instructor can take any number of courses
- A course can be taken by only one instructor
- A student can enrol for any number of courses
- Each course can have any number of students

